Arjuna JEE 2027

JEE MAINS & IIT JEE ADVANCED

MATHEMATICS

PERMUTATION & COMBINATION

DIBY (DO IT BY YOURSELF)

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DIBY-01

A man has 3 jackets, 10 shirts and 5 pairs of slacks.

If an outfit consists of a jacket, a shirt, and a pair of slacks, how many different outfits can the man make?

[Ans. : 150]

DIBY-02

There are 6 roads between A & B and 4 roads between B & C.

(i) In how many ways can one drive from A to C by way of B?

(ii) In how many ways can one drive from A to C and back to A, passing through B on both trips?

(iii) In how many ways can one drive the circular trip described in (ii) without using the same road more than once.

[Ans. : (i) 24; (ii) 576; (iii) 360]

DIBY-03

If repetitions are not permitted

(i) How many 3 digit numbers can be formed from the six digits 2, 3, 5, 6, 7 & 9?

(ii) How many of these are less than 400?

(iii) How many are even?

(iv) How many are odd?

(v) How many are multiple of 5?

[Ans. : (i) 120; (ii) 40; (iii) 40; (iv) 80; (v) 20]

DIBY-04

How many natural numbers are their from 1 to 1000 which have none of their digits repeated.

[Ans. : 738]

DIBY-05

How many positive integers less than 10^5 can be written with only the digits 7 and 9 ?

[Ans. : 62]

DIBY-06

(a) How many new words can be formed by using all the letters of the word 'MIRACLE'?

(b) How many words starts with a vowel and ending with a consonant.

(c) How many words starts with a consonant and ends with consonant.

[Ans. : (a) 7! – 1; (b) 1440; (c) 1440]

DIBY-07

Find the number of 10-digit numbers such that the product of any two consecutive digits in the number is a prime number.

[Ans. : 2048]

DIBY-08

Find number of ways in which 7 different colours in a rainbow can be arranged, if green is always in the middle.

[Ans. : 720]

DIBY-09

Find the number of natural numbers between 100 and 1000 such that atleast one of their digits is 7 .

[Ans. : 252]

DIBY-10

If the digits are not allowed to repeat in any numbers formed by using the digits 0, 2, 4, 6, 8, then the number of all numbers greater than 10,000 is equal to

[JEE Mains-2021]

[Ans. : 96]

DIBY-11

Find the number of 4 digit integers in the closed interval [2022, 4482] which can be formed using the digits {0, 2, 3, 4, 6, 7}. [JEE (Adv.)-2022]

[Ans. : 569]

DIBY-12

The number of 3-digit numbers, formed using the digits 2, 3, 4, 5 and 7, when the repetition of digits is not allowed, and which are not divisible by 3, is equal

[JEE Mains-2024]

[Ans. : 36]

DIBY-13

Total numbers of 3-digit numbers that are divisible by 6 and can be formed by using the digits 1, 2, 3, 4, 5 with repetition, is _____. [JEE Mains-2023]

[Ans. : 16]

DIBY-14

The number of words that can be formed out of the letters of the word ARTICLE so that the vowels occupy even places is

(A) 36 (B) 574 (C) 144 (D) 754

[Ans. : (C)]

DIBY-15

Eleven books consisting of 5 Mathematics, 4 Physics and 2 Chemistry are placed on a shelf. The number of possible ways of arranging them on the assumption that the books of the same subject are all together is

(A) 4!2! (B) 11! (C) 5!4!3!2! (D) None of these

[Ans. : (C)]

DIBY-16

The number of six letter words (with or without meaning), formed using all the letters of the word 'VOWELS', so that all the consonants never come together, is

(A) 576 (B) 140 (C) 356 (D) 209

[Ans. : (A)]

DIBY-17

Six married couple are sitting in a room. Then, the number of ways in which 4 people can be selected, so that They do not form a couple, is

(A) 240 (B) 255 (C) 480 (D) 495

[Ans. : (A)]

DIBY-18

How many of the arrangements of the letter of the word "LOGARITHM" begin with a vowel and end with a consonant?

[Ans. : 90720]

DIBY-19

Number of permutations of 1, 2, 3, 4, 5, 6, 7, 8 and 9 taken all at a time, such that the Digit 1 appearing somewhere to the left of 2
3 appearing to the left of 4 and
5 somewhere to the left of 6, is
(e.g. 815723946 would be one such permutation)

(A) $9 \cdot 7!$ (B) $8!$ (C) $5! \cdot 4!$ (D) $8! \cdot 4!$

[Ans. : (A)]

DIBY-20

A test consists of 6 multiple choice questions, each having 4 alternative answers of which only one is correct. The number of ways, in which a candidate answers all six questions such that exactly four of the answers are correct, is

[JEE Mains-2020]

[Ans. : 135]

DIBY-21

Let the number of elements in sets A and B be five and two respectively. Then the number of subsets of A × B each having at least 3 and at most 6 elements is

[JEE Mains-2023]

(A) 782 (B) 772 (C) 752 (D) 792

[Ans. : (D)]

DIBY-22

5 Indian & 5 American couples meet at a party & shake hands. If no wife shakes hands with her own husband & no Indian wife shakes hands with a male, then the number of hands shakes that takes place in the party is

(A) 95 (B) 110 (C) 135 (D) 150

[Ans. : (C)]

DIBY-23

Number of natural numbers between 100 and 1000 such that at least one of their digits is 7, is

(A) 225 (B) 243 (C) 252 (D) none

[Ans. : (C)]

DIBY-24

Let S = {1, 2, 3, ...9} for k = 1, 2, ...5. Let N_k be the number of subsets of S, each containing five elements out of which exactly k are odd.

Then $N_1 + N_2 + N_3 + N_4 + N_5$ = **[JEE (Adv.)-2017]**

(A) 125 (B) 210 (C) 252 (D) 126

[Ans. : (D)]

DIBY-25

The number of seven-digit numbers, that can be formed by using the digits 1, 2, 3, 5 and 7 such that each digit is used at least once. is:

[JEE Mains-2026]

(A) 15400 (B) 17800 (C) 16800 (D) 29400

[Ans. : (C)]

DIBY-26

Let P_n denote the total number of triangles formed by joining the vertices of an n-side regular polygon. If $P_{n+1} - P_n = 66$, then the sum of all distinct prime divisors of n is:

[JEE Mains-2026]

(A) 7 (B) 8 (C) 5 (D) 6

[Ans. : (C)]

DIBY-27

The number of ways, of forming a queue of 4 boys and 3 girls such that all the girls are not together, is:

[JEE Mains-2026]

(A) 5040 (B) 3050 (C) 3410 (D) 4320

[Ans. : (D)]

DIBY-28

A box contains 5 blue, 6 yellow and 4 red balls The number of ways, of drawing 8 balls containing at least two bails of each colour, is:

[JEE Mains-2026]

(A) 4100 (B) 4140 (C) 4230 (D) 4290

[Ans. : (A)]

DIBY-29

The number of numbers greater than 5000, less than 9000 and divisible by 3, that can be formed using the digits 0, 1, 2, 5, 9, if the repetition of the digits is allowed, is

[JEE Mains-2026]

[Ans. : 42]

DIBY-30

Let $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$. Let x be the number of 9-digit numbers formed using the digits of the set S such that only one digit is repeated and it is repeated exactly twice. Let y be the number of 9-digit numbers formed using the digits of the set S such that only two digits are repeated and each of these is repeated exactly twice. Then, **[JEE Mains-2026]**

(A) $56x = 9y$ **(B)** $29x = 5y$ **(C)** $45x = 7y$ **(D)** $21x = 4y$

[Ans. : (D)]

DIBY-31

There are 12 points in a plane, no three of which are in the same straight line, except 5 points which are collinear. Then the total number of triangles that can be formed with the vertices at any three of these 12 points is

[JEE Mains-2025]

(A) 230 **(B)** 220 **(C)** 200 **(D)** 210

[Ans. : (D)]

DIBY-32

The number of five digit numbers, greater than 40000 and divisible by 5, which can be formed using the digits 0, 1, 3, 5, 7 and 9 without repetition, is equal to **[JEE Mains-2023]**

(A) 120 (B) 132 (C) 72 (D) 96

[Ans. : (A)]

DIBY-33

The total number of three-digit numbers, divisible by 3, which can be formed using the digits 1, 3, 5, 8, if repetition of digits is allowed, is: **[JEE Mains-2023]**

(A) 22 (B) 18 (C) 21 (D) 20

[Ans. : (A)]

DIBY-34

If $^{2n+1}P_{n-1} : {}^{2n-1}P_n = 11:21$**, then** $n^2 + n + 15$ **is equal to:** **[JEE Mains-2023]**

[Ans. : 45]

DIBY-35

A person forgets his 4-digit ATM pin code. But he remembers that in the code all the digits are different, the greatest digit is 7 and the sum of the first two digits is equal to the sum of the last two digits. Then the maximum number of trials necessary to obtain the correct code is

[JEE Mains-2023]

[Ans. : 72]

DIBY-36

The number of seven digit positive integers formed using the digits 1, 2, 3 and 4 only and sum of the digits equal to 12 is _____. **[JEE Mains-2023]**

[Ans. : 413]

DIBY-37

A boy needs to select five courses from 12 available courses, out of which 5 courses are language courses. If he can choose at most two language courses, then the number of ways he can choose five courses is **[JEE Mains-2023]**

[Ans. : 546]

DIBY-38

There are 5 points P_1, P_2, P_3, P_4, P_5 on the side AB, excluding A and B, of a triangle ABC. Similarly there are 6 points $P_6, P_7, \ldots, P_{11}$ on the side BC and 7 points $P_{12}, P_{13}, \ldots, P_{18}$ on the side CA of the triangle.

The number of triangles, that can be formed using the points $P_1, P_2, \ldots, P_{18}$ as vertices, is: [JEE Mains-2024]

(A) 751 (B) 776 (C) 796 (D) 771

[Ans. : (A)]

DIBY-39

A committee of 5 is to be chosen from a group of 9 people. Number of ways in which it can be formed if two particular persons either serve together or not at all and two other particular persons refuse to serve with each other, is

(A) 41 (B) 36 (C) 47 (D) 76

[Ans. : (A)]

DIBY-40

There are 15 players in a cricket team, out of which 6 are bowlers, 7 are batsmen and 2 are wicketkeepers. The number of ways, a team of 11 players be selected from them so as to include at least 4 bowlers, 5 batsmen, and 1 wicketkeeper, is

[Ans. : 777]

DIBY-41

Suppose that 20 pillars of the same height have been erected-along the boundary of a circular stadium. If the top of each pillar has been connected by beams with the top of all its nonadjacent pillars, then the total number of beams is:

(A) 190 (B) 170 (C) 180 (D) 210

[Ans. : (B)]

DIBY-42

Let $A = \{1, 2, 3, 4, 5, 6, 7\}$ and $B = \{3, 6, 7, 9\}$. Then the number of elements in the set $\{C \subseteq A : C \cap B \neq \phi\}$ is_____.

[JEE Mains-2022]

DIBY-43

If ${}^{2n}C_3 : {}^{n}C_3 = 10 : 1$, then the ratio $(n^2 + 3n) : (n^2 - 3n + 4)$ is

[JEE Mains-2023]

(A) 65 : 37 (B) 2 : 1 (C) 36 : 16 (D) 27 : 11

[Ans. : (B)]

DIBY-44

If ${}^{n}P_r = {}^{n}P_{r+1}$ and ${}^{n}C_r = {}^{n}C_{r-1}$, then the value of r is equal to [JEEE Mains-2021]

(A) 1 (B) 4 (C) 2 (D) 3

[Ans. : (C)]

DIBY-45 or 37

A boy needs to select five courses from 12 available courses, out of which 5 courses are language course. If he can choose at most two language courses, then the number of ways he can choose five courses is [JEE Mains-2023]

[Ans. : 546]

DIBY-46

Consider a rectangle ABCD having 5, 7, 6, 9 points in the interior of the line segments AB, CD, BC, DA respectively. Let α be the number of triangles having these points from all 4 different sides as vertices and β be the number of quadrilaterals having these points from all 4 different sides as vertices. Then $(\beta - \alpha)$ is equal to [JEE Mains-2021]

(A) 795 (B) 1173 (C) 1890 (D) 717

[Ans. : (D)]

DIBY-47

There are 4 men and 5 women in Group A, and 5 men and 4 women in Group B. If 4 persons are selected from each group, then the number of ways of selecting 4 men and 4 women is _____. [JEE Mains-2024]

[Ans. : 5626]

DIBY-48

The letters of the word OUGHT are written in all possible ways and these words are arranged as in a dictionary, in a series. Then the serial number of the word TOUGH is [JEE Mains-2023]

(A) 89 (B) 84 (C) 86 (D) 79

[Ans. : (A)]

DIBY-49

All words, with or without meaning, are made using all the letters of the word MONDAY. These words are written as in a dictionary with serial numbers. The serial number of the word MONDAY is [JEE Mains-2023]

(A) 327 (B) 328 (C) 324 (D) 326

[Ans. : (A)]

DIBY-50

All the letters of the word PUBLIC are written in all possible orders and these words are written as in a dictionary with serial numbers. Then the serial number of the word PUBLIC is [JEE Mains-2023]

(A) 576 (B) 578 (C) 580 (D) 582

[Ans. : (D)]

DIBY-51

The letters of the word 'MANKIND' are written in all possible orders and arranged in serial order as in an English dictionary. Then the serial number of the word 'MANKIND' is [JEE Mains-2022]

[Ans. : 1492]

DIBY-52

All the letters of the word "GTWENTY" are written in all possible ways with or without meaning and these words are written as in a dictionary. The serial number of the word "GTWENTY" is [JEE Mains-2024]

[Ans. : 553]

DIBY-53

60 words can be made using all the letters of the word BHBJO, with or without meaning. If these words are written as in a dictionary, then the 50th word is: [JEE Mains-2024]

(A) JBBOH (B) OBBJH (C) OBBHJ (D) HBBJO

[Ans. : (B)]

DIBY-54

If all the words with or without meaning made using all the letters of the word "NAGPUR" are arranged as in a dictionary, then the word at 315th position in this [JEE Mains-2024]

(A) NRAGUP (B) NRAPUG (C) NRAPGU (D) NRAGPU

[Ans. : (C)]

DIBY-55

Find the total number of ways of selecting five letters from the work 'INDEPENDENT'.

[Ans. : 72]

DIBY-56

The number of ways five alphabets can be chosen from the alphabets of the word MATHEMATICS, where the chosen alphabets are not necessarily distinct, is equal to [JEE Mains-2024]

(A) 179 (B) 177 (C) 181 (D) 175

[Ans. : (A)]

DIBY-57

The total number of words (with or without meaning) that can be formed out of the letters of the word 'DISTRIBUTION' taken four at a time, is equal to [JEE Mains-2024]

[Ans. : 3734]

DIBY-58

In an examination of Mathematics paper, there are 20 questions of equal marks and the question paper is divided into three sections: A, B and C. A student is required to attempt total 15 questions taking at least 4 questions from each section. If section A has 8 questions, section B has 6 questions and section C has 6 questions, then the total number of ways a student can select 15 questions is [JEE Mains-2024]

[Ans. : 11376]

DIBY-59

In how many ways 5 NEET & 5 JEE Aspirants can be seated

(a) All JEE aspirants are together

(b) All NEET aspirants are together & all JEE aspirants are together

(c) Not all JEE aspirants are together

(d) No 2 JEE aspirants are together.

[Ans. : .....]

DIBY-60

Find the number of all 6 digit numbers such that all the digits of each number are selected from the set {1, 2, 3, 4, 5} and any digit that appears in the number appears at least twice.

[Ans. : 1405]

DIBY-61

The number of words that can be formed using all the letter of the word "CURRICULUM" in which two U's are together but separated from the third 'U' in $K(7!)$. The value of K is

(A) 7 (B) 9 (C) 12 (D) 14

[Ans. : (D)]

DIBY-62

The total number of ways in which 5 balls of different colours can be distributed among 3 persons so that each person gets at least one ball is [IIT-JEE-2012]

(A) 75 (B) 150 (C) 210 (D) 243

[Ans. : (B)]

DIBY-63

The students $S_1, S_2, \ldots, S_{10}$ are to be divided into 3 groups A, B and C such that each group has at least one students and the group C has at most 3 students. Then the total number of possibilities of forming such groups is [JEE Mains-2021]

[Ans. : 31650]

DIBY-64

The total number of three-digit numbers, divisible by 3, which can be formed using the digits 1, 3, 5, 8 if repetition of digits is allowed, is [JEE Mains-2023]

(A) 21 (B) 22 (C) 18 (D) 20

[Ans. : (B)]

DIBY-65

Let the set S = {2, 4, 8, 16, ... 512} be partitioned into 3 sets A, B, C with equal number of elements such that $A \cup B \cup C = S$ and $A \cap B = B \cap C = A \cap C = \phi$. The maximum number of such possible partitions of S is equal to [JEE Mains-2024]

(A) 1640 (B) 1520 (C) 1710 (D) 1680

[Ans. : (D)]

DIBY-66

A natura number has prime factorization given by $n = 2^x 3^y 5^z$, where y and z are such that $y + z = 5$ and $y^{-1} + z^{-1} = \frac{5}{6}, y > z.$ Then the number of odd divisors of n, including 1, is [JEE Mains-2021]

(A) 11 (B) $6x$ (C) 12 (D) 6

[Ans. : (C)]

DIBY-67

Number of integral solutions to the equation $x + y + z = 21$, where $x \geq 1, y \geq 3, z \geq 4$, is equal to [JEE Mains-2023]

[Ans. : 105]

DIBY-68

There are infinite number of Identical BLUE, RED, GREEN and YELLOW balls available for selections. In how many ways can we select 10 balls.

(A) $^{14}C_3$ (B) $^{13}C_{10}$ (C) $^{12}C_{10}$ (D) $^{14}C_4$

[Ans. : (B)]

DIBY-69

The number of triangles whose vertices are at the vertices of a regular octagon but none of whose sides is a side of the octagon is

[JEE Mains-2024]

(A) 56 (B) 16 (C) 24 (D) 48

[Ans. : (B)]

DIBY-70

The number of ways, 16 identical cubes, of which 11 are blue and rest are red, can be placed in row so that between any two red cubes, there should be at least 2 blue cubes, is

[JEE Mains-2022]

[Ans. : 56]

DIBY-71

Number of ways of arranging 8 identical books into 4 identical shelves where any number of shelves may remain empty is equal to

[JEE Mains-2024]

(A) 18 (B) 16 (C) 12 (D) 15

[Ans. : (D)]

DIBY-72

The total number of positive integral solutions (x, y, z) such that $xyz = 24$ is **[JEE Mains-2021]**

(A) 36 **(B) 45** **(C) 24** **(D) 30**

[Ans. : (D)]

DIBY-73

Let $\alpha = \frac{(4!)!}{(4!)^{3!}}$ and $\beta = \frac{(5!)!}{(5!)^{4!}}$, then **[JEE Mains-2024]**

(A) $\alpha \in$ **N and** $\beta \in$ **N**

(B) $\alpha \in$ **N and** $\beta \notin$ **N**

(C) $\alpha \notin$ **N and** $\beta \in$ **N**

(D) $\alpha \notin$ **N and** $\beta \notin$ **N**

[Ans. : (A)]

DIBY-74

From a group of 7 batsmen and 6 bowlers, 10 players are to be chosen for a team, which should include atleast 4 batsmen and atleast 4 bowlers. One batsmen and one bowler who are captain and vice-captain respectively of the team should be included. Then the total number of ways such a selection can be made, is [JEE Mains-2025]

(A) 165 (B) 155 (C) 145 (D) 135

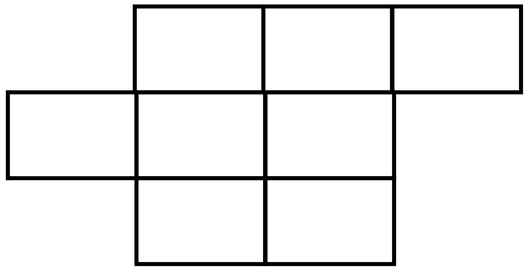
[Ans. : (B)]

DIBY-75

The number of ways, in which the letters A, B, C, D, E can be placed in the 8 boxes of the figure below so that no row remains empty and at most one letter can be placed in a box, is [JEE Mains-2025]

(A) 5880 (B) 960 (C) 840 (D) 5760

[Ans. : (D)]



DIBY-76

The number of functions f from {1, 2, 3, ..., 20} only {1, 2, 3, ..., 20} such that $f(k)$ is a multiple of 3, whenever k is a multiple of 4, is

[JEE Mains-2019]

(A) $6^5 \times (15)!$ (B) $5! \times 6!$ (C) $(15)! \times 6!$ (D) $5^6 \times 15$

[Ans. : (C)]

DIBY-77

The number of ways, 16 identical cubes, of which 11 are blue and rest are red, can placed in a row so that between any two red cubes there should be at least 2 blue cubes, is

[JEE Mains-2022]

[Ans. : 56]

DIBY-78

For $n \geq 2$, let S_n denote the set of all subsets of $\{1, 2, ..., n\}$ with no two consecutive numbers. For example $\{1, 3, 5\} \in S_6$, but $\{1, 2, 4\} \notin S_6$. Then $n(S_5)$ is equal to

[JEE Mains-2025]

[Ans. : 13]

DIBY-79

The largest $n \in N$ such that 3^n divides 50! is

[JEE Mains-2025]

(A) 21 (B) 22 (C) 20 (D) 23

[Ans. : (B)]

DIBY-80

In an examination 5 students have been allotted their seats as per their roll numbers. The number of ways, in which none of the students sits on the allotted seat, is

[JEE Mains-2023]

[Ans. : 44]

It is never about end result

It's always about Journey

THANK You

#futureIITians